

Hector Lugo III

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EDUCATION

The University of Texas Rio Grande Valley

Master of Science in Computer Science

Aug. 2024 – Dec. 2026

GPA: 3.79/4.00

The University of Texas Rio Grande Valley

Bachelor of Science in Computer Science

Aug. 2020 – May 2024

EXPERIENCE

AI/ML Research Intern

Air Force Research Laboratory (AFRL), Rome, NY

May 2026 – Present

- Developing multi-agent, multi-camera perception systems that coordinate visual understanding across multiple cameras and autonomous agents.
- Investigating deepfake detection and generation alongside ongoing robotics and reinforcement learning research for autonomous systems.

Machine Learning Engineer Intern

Idaho National Laboratory (INL)

May 2025 – Jun. 2026

- Engineered a ResNet autoencoder to denoise TDMS signals, significantly improving data quality for smart grid anomaly detection.
- Accelerated model training by over 93% (from 30 days to 1–2 days) using FFT preprocessing and HPC (Slurm).

Research Assistant

Machine Intelligence Lab, UTRGV

Jan. 2023 – Present

- Conducting Vision-Language Model (VLM) and Vision-Language-Action (VLA) research for humanoid locomotion and control using LocoMujoco.
- Built custom Isaac Sim environments to train scalable reinforcement learning policies for Unitree Go2 autonomous navigation.

Graduate Teaching Assistant

UTRGV

Jan. 2024 – Present

- Instruct CS 1101 (Intro to CS) and Digital Image Processing, mentoring 40+ students across lectures, labs, and grading.

Summer Research Intern

University of California Riverside

June 2023 – Aug. 2023

- Applied spectral clustering to U.S. railway accident data, identifying critical spatial safety zones and improving predictive risk assessment accuracy.
- Published findings on spatial safety analysis at the ASME JRC2024 Conference.

PROJECTS

Personalized Chemotherapy Dosing | *Decision Transformer, Offline RL*

Jun. 2024 – Jul. 2024

- Developed a Decision Transformer for personalized dose control using state-conditioned inputs.
- Accepted at ICICT 2025 for contributions to adaptive treatment planning via offline reinforcement learning.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, JavaScript, HTML/CSS, R

Frameworks & Tools: PyTorch, NumPy, Pandas, FastAPI, React, Flask, Git, Docker, Linux, HPC (Slurm)

Robotics & Simulation: Omniverse Isaac Lab, Isaac Sim, MuJoCo, LocoMujoco, ROS, Unitree Go2 Ecosystem

Concepts: Reinforcement Learning, VLM, VLA, Computer Vision, Signal Processing, Autoencoders, FFT, Decision Transformers